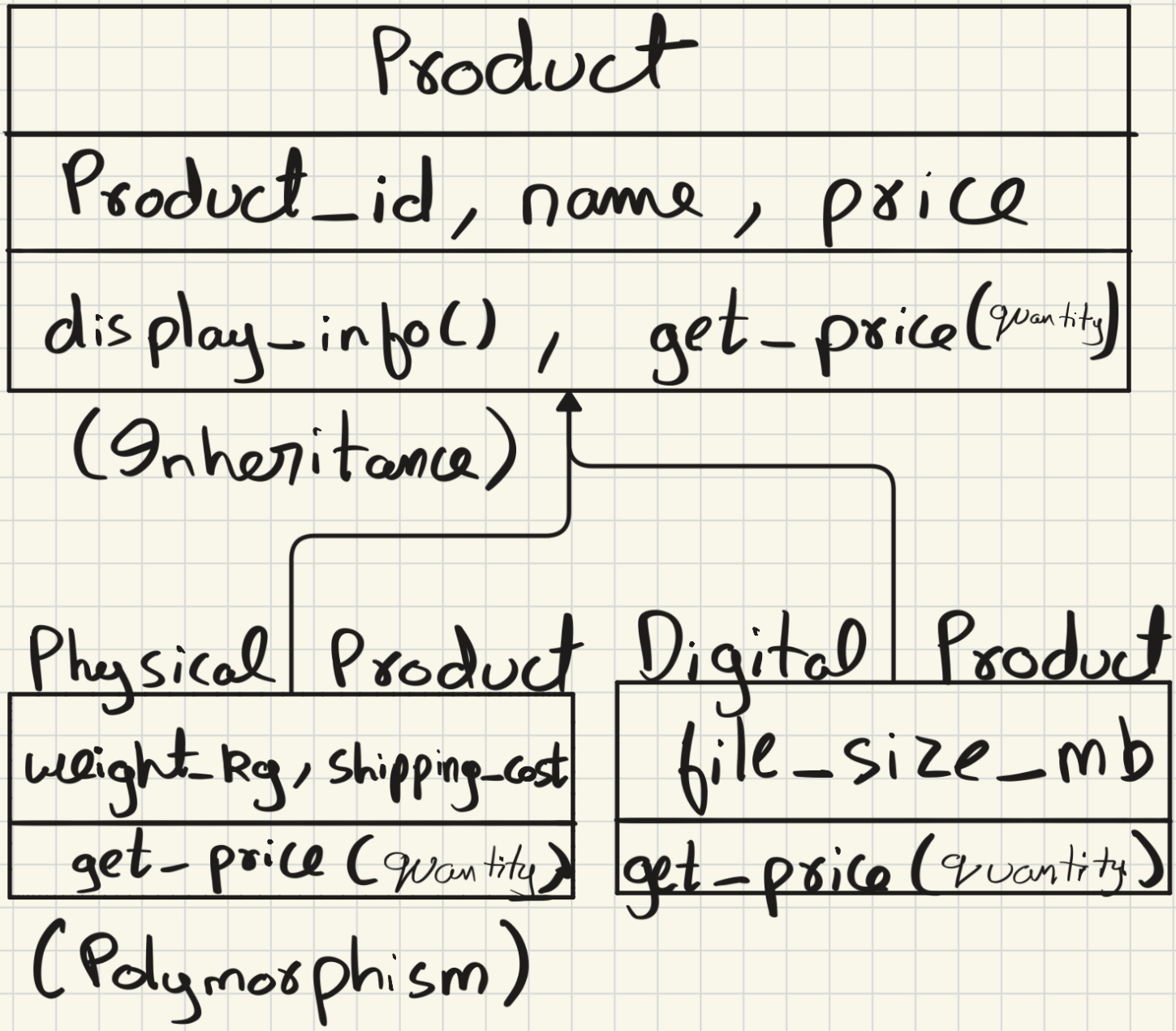
UML Diagram


Product
Product_id, name, price
display_info(), get_price(quantity)
(Inheritance)
Physical Product
weight_kg, shipping_cost
get_price(quantity)
Digital Product
file_size_mb
get_price(quantity)
(Polymorphism)

User Manual: Python E-Commerce System

1. Introduction

The Python E-Commerce program is a text-based shopping cart simulator. It demonstrates Object-Oriented Programming (OOP) by handling two distinct product types:

- **Physical Products:** Require shipping (adds weight-based costs).
- **Digital Products:** Downloadable files (no shipping, but bulk discounts apply).

2. System Requirements

- **Software:** Python 3.6 or higher.
- **Files:** Ensure shop.py and test_shop.py are in the same folder.

3. Step-by-Step Usage Guide

Step 1: View the Catalogue

- Select Option 1 from the main menu.
- The screen will display all available items, along with their corresponding **Product IDs** (e.g., 101, 201).

Step 2: Add Items to Cart

- Select Option 2.
- **Enter Product ID:** Type the ID of the item you wish to buy.
- **Enter Quantity:** Type a positive number.
- *Error Handling:* If a negative number or text is provided in the input, the system will show an error message and return to the menu.

Step 3: Checkout

- Select Option 3 to finish shopping.
- The system will print a receipt showing the breakdown of costs:
- **Physical Items:** Shows base price + calculated shipping.
- **Digital Items:** Shows base price - bulk discount (if you bought 5 or more).

The program will display the **Grand Total** and exit.

4. Troubleshooting

- **"Invalid Input":** Ensure you are typing numbers (1, 2, 3) for menu choices and quantities.
- **Program Closes Immediately:** If running from a file explorer, try running it from an IDE (like VS Code or PyCharm) or the terminal to keep the window open.

```
Select: 4
anirudh@Anirudhs-MacBook-Air ~ % /us
..
----------------------------------------
Ran 2 tests in 0.000s

OK
anirudh@Anirudhs-MacBook-Air ~ %
```